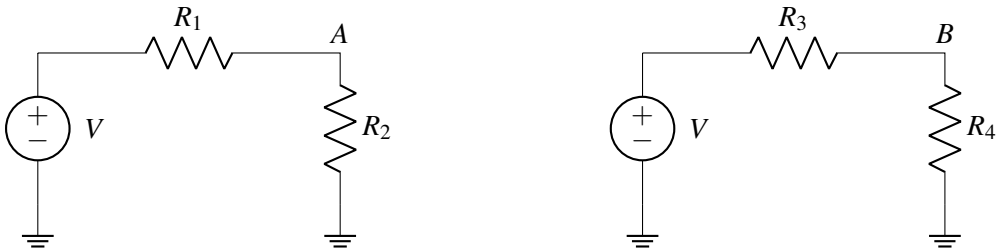
CSM
Spring 2026

EECS 16B
# Worksheet 3

## 1. Modular Circuits

In this question we will explore how to look at circuits in a 'modular' way. That is, we will start to encapsulate circuit components (just like one might encapsulate code into functions) and then see how those components fit together.

(a) Here, you can see two standard voltage dividers. Find the voltage at the points $A$ and $B$ using standard nodal analysis techniques.




**Solution:** We can apply KCL at node $A$. We get:

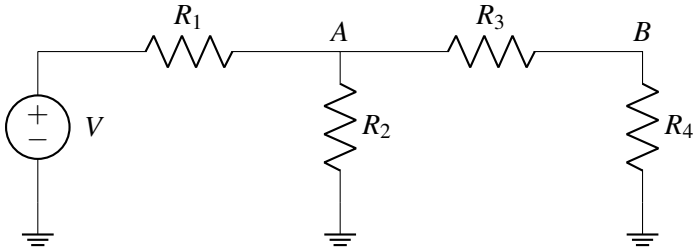
$$\frac{V_A - 0}{R_2} = \frac{V - V_A}{R_1}$$
$$\implies \frac{R_1}{R_2} V_A = V - V_A$$
$$\implies V_A\left(1 + \frac{R_1}{R_2}\right) = V$$
$$\implies V_A \frac{R_1 + R_2}{R_2} = V$$
$$\implies V_A = V \frac{R_2}{R_1 + R_2}$$

Similarly, for the second circuit, we have

$$V_B = V \frac{R_4}{R_3 + R_4}$$

(b) Now, let us modify this circuit to add a second voltage divider stage starting at A. You can think of this as cascading 2 voltage dividers: one which which has an input of $V$ and an output of $V_A$, and a second one that has an input of $V_A$ and an output of $V_B$. Find the voltage at $B$ using nodal analysis.

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**Solution:** We can apply KCL at node $A$ and at node $B$. We get the equations:

$$\frac{V-V_A}{R_1}+\frac{0-V_A}{R_2}=\frac{V_A-V_B}{R_3}$$
$$\frac{V_A-V_B}{R_3}=\frac{V_B-0}{R_4}$$

We can simplify these equations to give us

$$\frac{V}{R_1}-\frac{V_A}{R_1}-\frac{V_A}{R_2}=\frac{V_A}{R_3}-\frac{V_B}{R_3}$$
$$\implies \frac{V_B}{R_3}+\frac{V}{R_1}=\frac{V_A}{R_1}+\frac{V_A}{R_2}+\frac{V_A}{R_3}$$
$$\implies \frac{V_B}{R_3}+\frac{V}{R_1}=V_A(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3})$$
$$\frac{V_A}{R_3}-\frac{V_B}{R_3}=\frac{V_B}{R_4}$$
$$\implies \frac{V_A}{R_3}=\frac{V_B}{R_3}+\frac{V_B}{R_4}$$
$$\implies V_A=V_BR_3(\frac{1}{R_3}+\frac{1}{R_4})$$

Combining these 2 equations, we get:

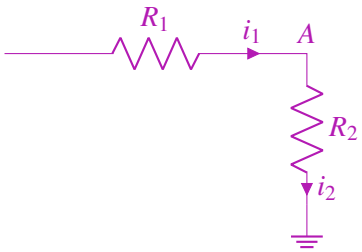
$$\frac{V_B}{R_3}+\frac{V}{R_1}=V_BR_3(\frac{1}{R_3}+\frac{1}{R_4})(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3})$$
$$\implies \frac{V_B}{R_3}+\frac{V}{R_1}=V_BR_3(\frac{1}{R_3}+\frac{1}{R_4})(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3})$$
$$\implies \frac{V}{R_1}=V_BR_3(\frac{1}{R_3}+\frac{1}{R_4})(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3})+\frac{V_B}{R_3}=V_B(R_3(\frac{1}{R_3}+\frac{1}{R_4})(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3})+\frac{1}{R_3})$$
$$\implies V_B=\frac{V}{R_1}/(R_3(\frac{1}{R_3}+\frac{1}{R_4})(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3})+\frac{1}{R_3})$$

(c) Compare your solwer with what you get when you multiplying the amplification factors of each individual voltage divider. Are they the same or different? Does this surprise you? Why or why not?

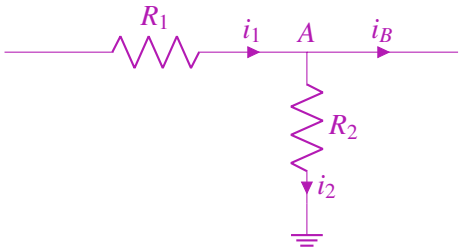
**Solution:** They are clearly not the same! The next part explains why.

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What we see here is called "loading". Ideally, we would like to be able to stack voltage dividers like this in order to cascade their effects. However this does not work because by adding the second circuit, we are changing what the first one does. In particular, the second circuit will draw some current from the first circuit. We can see this difference if we zoom in on the node A. In part (a), we had the following picture:




And we were able to say that $i_1 = i_2$, because they are the only currents at node $A$. But after adding the second voltage divider stage, the picture now looks like this:



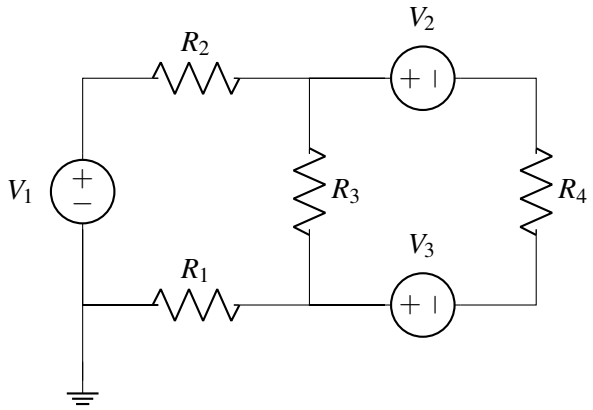

And so the KCL equation we wrote at node $A$ is no longer valid.

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## 2. Supernode

**Solution:** Prereq: Best placed after a super easy nodal analysis problem.
Description: This shows how to deal with multiple voltage sources that don't share a ground when doing nodal analysis since we don't know how much current goes through a voltage sources.

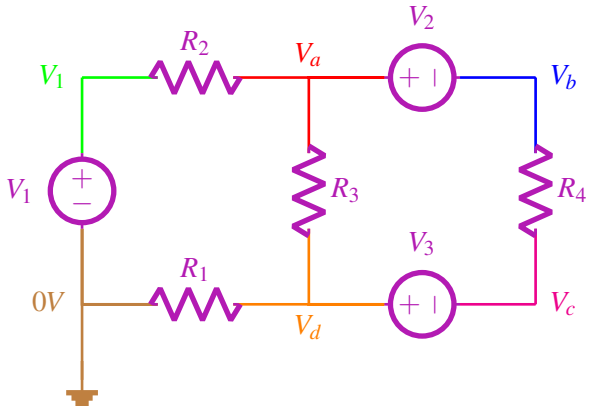
In this question, we will explore how to deal with multiple voltage sources when doing nodal analysis. Throughout the problem, assume that $R_1 = 1\Omega, R_2 = 2\Omega, R_3 = 4\Omega, R_4 = 4\Omega, V_1 = 1\text{V}, V_2 = 2\text{V}, V_3 = 4\text{V}$. While it is good practice to symbolically solve problems before substituting specific numbers, for superposition it is often easiest to substitute before superposing the solwers.




(a) Mark all the nodes. If you know the potential at the node, write down the value next to the node. If you don't know the value, then assign a variable for the potential.

**Solution:**

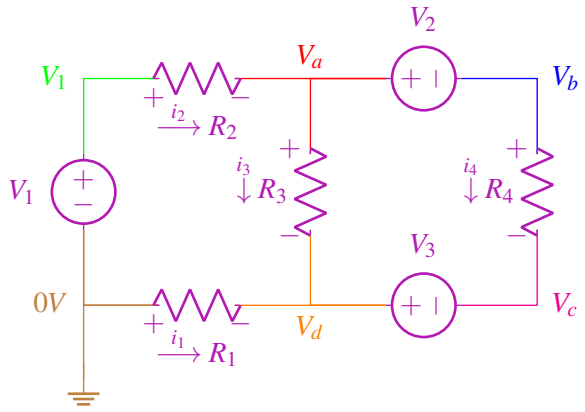
$V_a, V_b, V_c, V_d$ are variables. We know the potentials at the node marked ground and at the node marked $V_1$.




(b) Mark current directions arbitrarily and corresponding polarities on each resistor. Note that if current goes from left to right, then the left side of the resistor is to be marked + and the right side must be marked -. This is the passive sign convention.

**Solution:**

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**Solution:** Might be a good idea to mark the nodes in the same fashion as the solutions for later parts!

(c) Note that we define 4 nodes with unknown potentials. So we need 4 equations. Each of these nodes with unknown potential should give us one equation.
Write the equation for the first node with unknown potential.

**Solution:** For the $V_a$ node, we have a problem. The problem is that we don't know what current enters the voltage source $V_2$. The fix here is to treat $V_a$ and $V_b$ combined as a super-node. Since we are considering 2 nodes at the same time, this 'supernode' better give us 2 equations!
We can get equation from the fact that the voltage source exists.

$$V_a - V_b = V_2 \tag{1}$$

For the other equation, let's consider: (Sum of current entering nodes $V_a$ and $V_b$) = (Sum of current leading nodes $V_a$ and $V_b$).

**Consider $V_a$:**

- $i_2$ enters $V_a$.
- $i_3$ leaves $V_a$.

**Consider $V_b$:**

- $i_4$ leaves $V_b$.

$$i_2 = i_3 + i_4$$
$$\frac{V_1 - V_a}{R_2} = \frac{V_a - V_d}{R_3} + \frac{V_b - V_c}{R_4} \tag{2}$$

**Solution:** For equation 2, make sure students know that in the numerator, you have to write it as $V_+ - V_-$, depending on how you marked the polarities. But it is super important to mark the + where the current starts and - where it ends. This is passive sign convention. Make sure they do this right! Also, if students ask why (Sum of current entering nodes $V_a$ and $V_b$) = (Sum of current leading nodes $V_a$ and $V_b$) is something that we can use, show on the circuit that current cannot stay between $V_a$ and $V_b$. Any current that enters $V_a$ or $V_b$ must leave through $V_a$ or $V_b$

(d) Write equations for all remaining unknown nodes. Solve for all the unknown potentials.

**Solution:** Nodes $V_c$ and $V_d$ remain. Let's start at $V_c$. Again, we don't know what current enters $V_3$, so we do the same thing. Treat the 2 ends of the voltage source, $V_c$ and $V_d$ as a super-node. One equation we get is

$$V_d - V_c = V_3 \tag{3}$$

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The other equation comes from (Sum of current entering nodes $V_a$ and $V_b$) = (Sum of current leading nodes $V_a$ and $V_b$).

**Consider $V_c$:**

- $i_4$ enters.

**Consider $V_d$:**

- $i_3$ enters
- $i_1$ enters.

$$i_4 + i_3 + i_1 = 0$$
$$\frac{V_b - V_c}{R_4} + \frac{V_a - V_d}{R_3} + \frac{0 - V_d}{R_1} = 0 \tag{4}$$

Equations (1), (2), (3), (4) are four equations for the four variables $V_a, V_b, V_c, V_d$ and can be solved uniquely. One fun fact is that these equations are guaranteed to be linearly independent as long as the circuit is not an impossible circuit (such as 2 current sources of differing values in series, or 2 voltage sources of different values in parallel).

To summarize, we have:

$$\begin{cases} V_a - V_b = V_2 \\ \frac{V_1 - V_a}{R_2} = \frac{V_a - V_d}{R_3} + \frac{V_b - V_c}{R_4} \\ V_d - V_c = V_3 \\ \frac{V_b - V_c}{R_4} + \frac{V_a - V_d}{R_3} + \frac{0 - V_d}{R_1} = 0 \end{cases} \Longrightarrow \begin{cases} V_a - V_b = 2 \\ \frac{1 - V_a}{2} = \frac{V_a - V_d}{4} + \frac{V_b - V_c}{4} \\ V_d - V_c = 4 \\ \frac{V_b - V_c}{4} + \frac{V_a - V_d}{4} + \frac{-V_d}{1} \end{cases}$$

Solving the right set of equations, we find that $\boxed{V_a = \frac{1}{5}\text{V}, V_b = -\frac{9}{5}\text{V}, V_c = -\frac{18}{5}\text{V}, V_d = \frac{2}{5}\text{V}}$.

(e) Now we will explore solving the circuit using superposition. To recall the process, which 2 sources will be suppressed when looking at each of the 3 sources. Will the other 2 sources be treated as open circuits or short circuits?

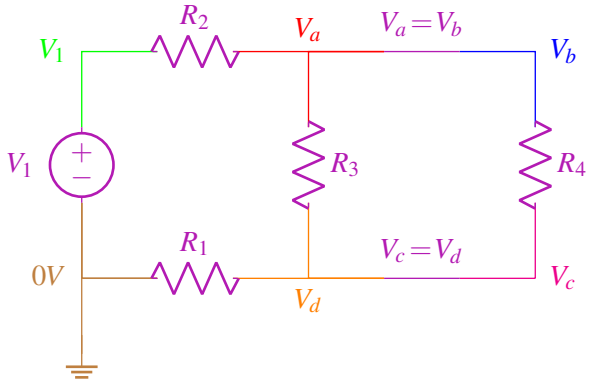
**Solution:** For $V_1$, $V_2$ and $V_3$ will be shorted. For $V_2$, $V_1$ and $V_3$ will be shorted. For $V_3$, $V_1$ and $V_2$ will be shorted. When doing superposition and suppressing inputs, a voltage source is treated as a short and a current source is treated as an open.

The reasoning for "zeroing" voltage/current sources is as follows: we define a voltage source as something that maintains a fixed voltage drop between the two terminals. If we "zero" that source, we are saying the voltage across the two terminals should be 0V. But that, by definition, has the same effect as connecting the two terminals with a wire, i.e. shorting it! So we say "zeroed" voltage sources are shorted. A similar argument leads us to find "zeroed" current sources are equivalent to open circuits.

(f) Draw the superposition circuit for $V_1$ and solve this circuit.

**Solution:**

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$$\begin{cases} \dfrac{V_a - V_1}{R_2} + \dfrac{V_a - V_d}{R_3} + \dfrac{V_b - V_c}{R_4} = 0 \\ \dfrac{V_d}{R_1} + \dfrac{V_d - V_a}{R_3} + \dfrac{V_c - V_b}{R_4} = 0 \\ V_a = V_b \\ V_c = V_d \end{cases} \quad \Longrightarrow \quad \begin{cases} \dfrac{V_a - V_1}{R_2} + \dfrac{V_a - V_d}{R_3} + \dfrac{V_a - V_d}{R_4} = 0 \\ \dfrac{V_d}{R_1} + \dfrac{V_d - V_a}{R_3} + \dfrac{V_d - V_a}{R_4} = 0 \end{cases}$$

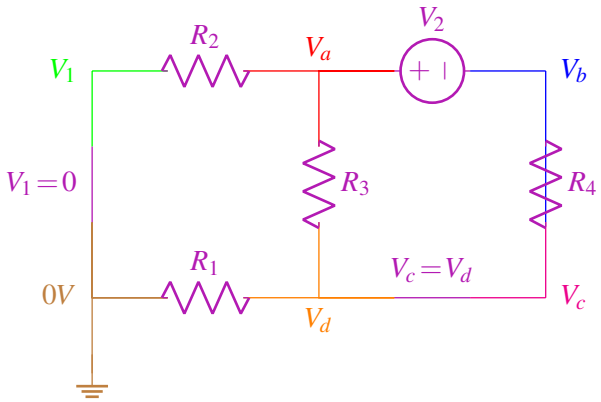
Solving for $V_a, V_b, V_c,$ and $V_d$ we find

$$\begin{cases} V_a = V_b = \dfrac{(R_1R_2 + R_1R_4 + R_3R_4)}{R} V_1 \\ V_d = V_c = \dfrac{R_1R_3 + R_1R_4}{R} V_1, \end{cases} \qquad R \equiv R_1R_3 + R_2R_3 + R_1R_4 + R_2R_4 + R_3R_4$$

Or, if you prefer solving with numbers first:

$$\begin{cases} \dfrac{V_a - 1}{2} + \dfrac{V_a - V_d}{4} + \dfrac{V_a - V_d}{4} = 0 \\ \dfrac{V_d}{1} + \dfrac{V_d - V_a}{4} + \dfrac{V_d - V_a}{4} = 0 \end{cases} \Longrightarrow \boxed{\begin{cases} V_a = \frac{3}{5} \\ V_b = \frac{3}{5} \\ V_c = \frac{1}{5} \\ V_d = \frac{1}{5} \end{cases}}$$

(g) Draw the superposition circuit for $V_2$ and write the equation that characterizes this circuit.
**Solution:**



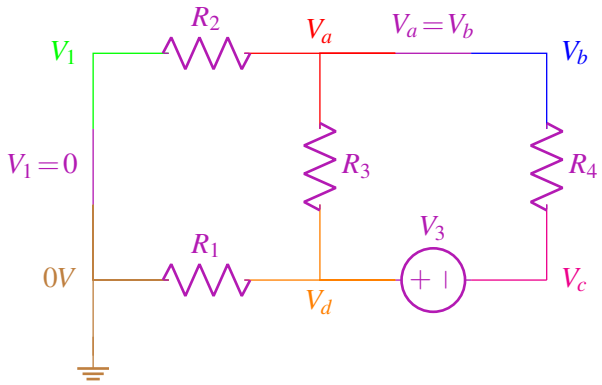

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$$\begin{cases} \dfrac{V_1 - V_a}{R_2} + \dfrac{V_1 - V_d}{R_1} = 0 \implies V_a = -2V_d \\ \dfrac{V_d}{R_1} + \dfrac{V_d - V_a}{R_3} + \dfrac{V_c - V_b}{R_4} = 0 \implies 5V_d - V_a + V_c - V_b = 0 \\ V_c = V_d \\ V_a = V_b + V_2 = V_b + 2 \end{cases}$$

$$\begin{cases} V_a = -2V_d \\ 6V_d - 2V_a + 2 = 0 \end{cases} \implies \boxed{\begin{cases} V_a = \dfrac{2}{5} \\ V_b = -\dfrac{8}{5} \\ V_c = -\dfrac{1}{5} \\ V_d = -\dfrac{1}{5} \end{cases}}$$

(h) Draw the superposition circuit for $V_3$ and write the equation that characterizes this circuit.

**Solution:**




$$\begin{cases} \dfrac{V_a - V_1}{R_2} + \dfrac{V_a - V_d}{R_3} + \dfrac{V_b - V_c}{R_4} = 0 \\ \dfrac{V_d}{R_1} + \dfrac{V_d - V_a}{R_3} + \dfrac{V_c - V_b}{R_4} = 0 \\ V_d = V_c + V_3 = V_c + 4 \\ V_a = V_b \\ V_1 = 0 \end{cases} \implies \begin{cases} \dfrac{V_a}{2} + \dfrac{V_a - V_d}{4} + \dfrac{V_b - V_c}{4} = 0 \\ V_d + \dfrac{V_d - V_a}{4} + \dfrac{V_c - V_b}{4} = 0 \\ V_d = V_c + 4V_a = V_b \end{cases}$$

We find that: $\boxed{V_a = -\dfrac{4}{5}\text{V}, V_b = -\dfrac{4}{5}\text{V}, V_c = -\dfrac{18}{5}\text{V}, V_d = \dfrac{2}{5}\text{V}}$

(i) Find the unknown potentials as you did in in part (d), but now do so using the superposition theorem.

**Solution:** By the superposition theorem, the actual potential values at nodes $V_a$ through $V_d$ are the linear sums of their corresponding potentials in the simpler circuits. Notice that we kept the ground

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in the same place for each of the simpler circuits! While the location of the ground does not matter, *keeping it consistent does*. We add up the potentials and find that:

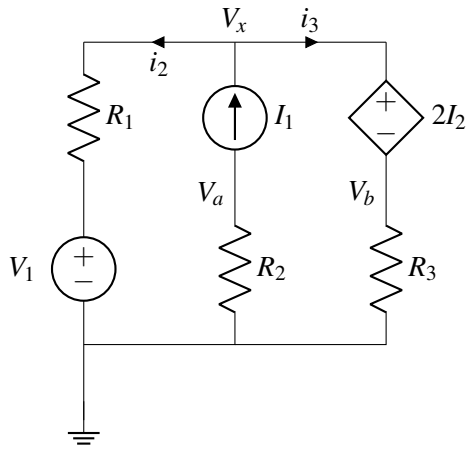
$$V_a = \frac{3}{5} + \frac{2}{5} - \frac{4}{5} = \frac{1}{5}$$
$$V_b = \frac{3}{5} - \frac{8}{5} - \frac{4}{5} = -\frac{9}{5}$$
$$V_c = \frac{1}{5} - \frac{1}{5} - \frac{18}{5} = -\frac{18}{5}$$
$$V_d = \frac{1}{5} - \frac{1}{5} + \frac{2}{5} = \frac{2}{5}$$

Wow! The same potentials as we found using supernode/nodal analysis!

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**3. It's a bird! It's a plane! No, it's superposition!**

(a) Solve the following circuit for $V_x$ using nodal analysis. Let $R_1 = 10\Omega$, $R_2 = 5\Omega$, $R_3 = 2\Omega$, $V_1 = 7V$, and $I_1 = 6A$.




Let us examine the case in which we see all currents flowing OUT of $V_x$:

For the current flowing through $R_1$, $I_2$, we need to use Ohm's law. We know easily that above $R_1$ the voltage is $V_x$, as it is connected to the same node. Beneath, we have a voltage gain of $V_1$ from ground. (Remember, voltage sources power voltage differences rather than absolute values). Thus we have

$$I_2 = \frac{V_x - V_1}{R_1}$$

Since we're setting all currents as going out of $V_x$, we'll say the current going out on the branch with $I_1$ as $-I_1$.

Finally, the branch with dependent sources. The easiest way to find the current, again, is using the resistor $R_3$ and finding the current going through it. While the bottom terminal is connected to ground, what is the top terminal? Say we followed the wire down from $V_x$. Then, when we cross the dependent voltage source, since sources power differences in voltage, we can say the voltage below it is equal to $V_b = V_x - 2I_2$.

Thus, current through $R_3$, $I_3$ is

$$I_3 = \frac{V_x - 2I_2}{R_3}$$

.

We have the sum of currents equal to 0, as they are all going out.

Thus,

$$I_2 + (-I_1) + I_3 = 0$$
$$\frac{V_x - V_1}{R_1} = I_1 - \frac{V_x - 2I_2}{R_3}$$

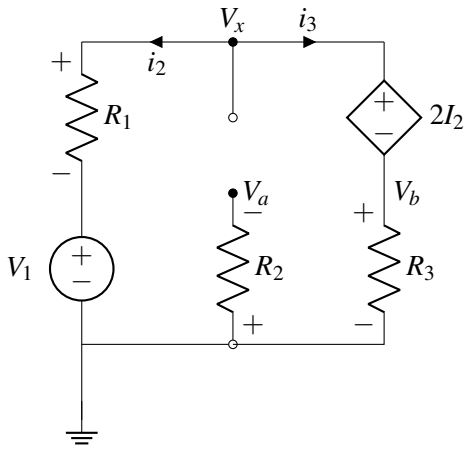
Where $I_2$ is known. Thus we can solve this equation! Plugging in numbers,

$$I_2 = \frac{V_x - 7}{10}$$

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$$\frac{V_x-7}{10}=6-\frac{V_x-2I_2}{2}$$
$$\frac{V_x-7}{10}=6-\frac{V_x-\frac{V_x-7}{5}}{2}$$
$$\frac{V_x-7}{5}=12-V_x-\frac{V_x-7}{5}$$
$$V_x=12$$

(b) Now solve the same circuit for $V_x$ using superposition.
Superposition circuit when removing the current source




Setting this up, we get three equations, for $V_a$ we get:

$$V_a=0$$

Next, lets observe the value of $I_2$ so that we can use that with the dependent source:

$$I_2=\frac{V_x-V_1}{R_1}$$

Using this, we can now express the difference of voltage over the dependent source, i.e.:

$$V_x-V_b=2I_2=2\frac{V_x-V_1}{R_1}$$

Next we note that in this version, that the current doesn't split anywhere, so $I_2=-I_3$. So, we can find the voltage drop over $R_3$ as follows:

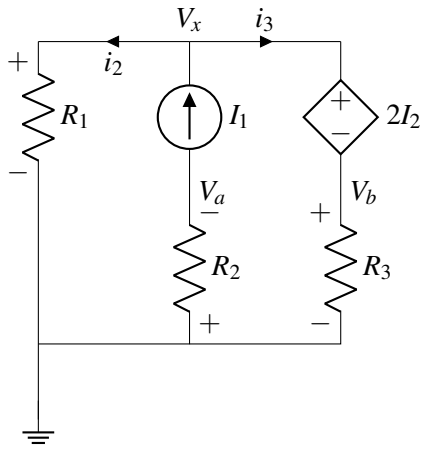
$$\frac{0-V_b}{R_3}=I_3=-I_2=-\frac{V_x-V_1}{R_1}$$

So, repeating the above, our three equations are:

$$V_a=0 \tag{5}$$
$$V_x-V_b=2\frac{V_x-V_1}{R_1} \tag{6}$$
$$\frac{0-V_b}{R_3}=-\frac{V_x-V_1}{R_1} \tag{7}$$

Superposition circuit while removing the voltage source

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Now, we will find the equivalent three equations for this form. Starting again with the voltage drop over $R_2$, we have:

$$\frac{0-V_a}{R_2} = I_1$$

Next, again, let's find $I_2$: $I_2 = \frac{V_x-0}{R_1} = \frac{V_x}{R_1}$ Using this, we can find the difference over the dependent source:

$$V_x - V_b = 2I_2 = 2\frac{V_x}{R_1}$$

Finally, we can express the voltage drop over $R_3$ via KCL:

$$I_1 = I_2 - I_3 = \frac{V_x}{R_1} - I_3 = \frac{V_x}{R_1} + \frac{0-V_b}{R_3}$$

We now again have three equations as follows:

$$\frac{0-V_a}{R_2} = I_1 \tag{8}$$

$$V_x - V_b = 2\frac{V_x}{R_1} \tag{9}$$

$$I_1 = \frac{V_x}{R_1} + \frac{0-V_b}{R_3} \tag{10}$$

Now, we just plug in the numbers we have into our 6 numbered equations, and add up the values for matching voltages, and as a result we get:

For nulling the current source:

$$-V_b = \frac{2(V_x-7)}{10}$$

$$V_x + \frac{2(V_x-7)}{10} = 2\frac{V_x-7}{10}$$

$$V_x = 0$$

For nulling the voltage source:

$$6 = \frac{V_x}{10} - \frac{V_b}{2}$$

$$-V_b = 12 - \frac{V_x}{5}$$

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$$V_x - 12 + \frac{V_x}{5} = 2\frac{V_x}{10}$$
$$V_x = 12$$

Then, adding the two values for $V_x$ together,

$$0 + 12 = 12$$

We have verified that the answer we get for superposition is the same as the answer we get using nodal analysis.

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